

Math Review

ECON 105: Intro to Mathematical Economics

Patrick Molligo

UCLA

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Agenda

- 1 Optimization
- 2 First & second order conditions
- 3 Lagrange multiplier method
- 4 Practice

How has your first week at UCLA been?

- A. Great! Classes are easy.
- B. Good, but some things are confusing.
- C. Feeling challenged.
- D. I'm completely lost.

Find the partial derivative $\frac{\delta f(x)}{\delta x_2}$:

$$f(x) = (5x_1^{1/3} + 3x_2^{1/3})^3$$

- A. $(5x_1^{1/3} + 3x_2^{1/3})^2(5x_1^{-2/3})$
- B. $(5x_1^{1/3} + 3x_2^{1/3})^2(3x_2^{-2/3})$
- C. $(5x_1^{1/3} + 3x_2^{1/3})^2(\frac{5}{3}x_1^{-2/3} + x_2^{-2/3})$
- D. $3(5x_1^{1/3} + 3x_2^{1/3})^2(\frac{5}{3}x_1^{-2/3} + x_2^{-2/3})$

What is optimization?

- In economics, assume consumers and firms make **optimal decisions**
 - Given preferences and budgets, how do buyers **maximize** their well-being (utility) and **minimize** their spending?
 - Given production capabilities, how do firms **maximize** their profits and **minimize** their costs?
- Optimization is all about maximizing and/or minimizing functions using calculus

How do we optimize?

- Big picture: if a function is still increasing, we haven't hit the maximum yet. When a function starts decreasing, we went past the maximum.
 - Reverse is true when finding the minimum
- At any value $x = c$

$$f'(c) > 0 \longrightarrow \text{increasing}$$

$$f'(c) < 0 \longrightarrow \text{decreasing}$$

$$f'(c) = 0 \longrightarrow \text{neither}$$

Critical points

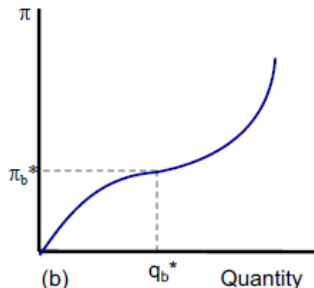
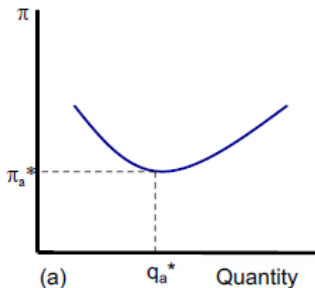
- If $f'(c) = 0$ we call c a **critical point**
 - Usually denote as x^*
- Common notation: $\left. \frac{df(x)}{dx} \right|_{x=x^*} = 0$
 - “The derivative of $f(x)$ evaluated at the point x^* equals 0”

Tip: try visualizing

- Graphing a function often clarifies things
- Useful online tool: <https://www.desmos.com/calculator>
 - Let's look at $f(x) = x^2 - 36x + 100$
 - Use the ^ symbol for exponents
 - What is the critical point? Maximum or minimum?
- OK to use technology as an aid, but it won't be there for exams!

First order condition

- The First Order Condition (FOC) for a maximum/minimum
 - $f'(x^*) = 0$
 - “**Necessary** but not **sufficient**”, why?



Note: We use π for firm profits (Greek “P”)

Second order condition

- Just because $f'(x^*) = 0$ doesn't mean we found the maximum/minimum
- Second Order Condition (SOC):

$$f''(x^*) < 0 \longrightarrow \text{(local) maximum}$$

$$f''(x^*) > 0 \longrightarrow \text{(local) minimum}$$

Second order condition

- Just because $f'(x^*) = 0$ doesn't mean we found the maximum/minimum
- Second Order Condition (SOC):

$$f''(x^*) < 0 \longrightarrow \text{(local) maximum}$$

$$f''(x^*) > 0 \longrightarrow \text{(local) minimum}$$

- At the top of a mountain, the only way to go is down
 - If slope is increasing, you're not at the top yet
 - Reverse is true at the bottom

Multivariable case

- Suppose we have 2 variables $f(x_1, x_2)$
 - FOC works exactly the same, just do it twice

$$\frac{\delta f(x_1, x_2)}{\delta x_1} = f_1 = 0$$

$$\frac{\delta f(x_1, x_2)}{\delta x_2} = f_2 = 0$$

- SOC has a new requirement. For a maximum:

$$\frac{\delta^2 f(x_1, x_2)}{\delta x_1^2} = f_{11} < 0$$

$$\frac{\delta^2 f(x_1, x_2)}{\delta x_2^2} = f_{22} < 0$$

$$f_{11}f_{22} - (f_{12})^2 > 0$$

Partial derivative recap: notation

- f_1 : derivative w.r.t. x_1 , hold x_2 constant
- f_2 : derivative w.r.t. x_2 , hold x_1 constant

Partial derivative recap: notation

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- f_2 : derivative w.r.t. x_2 , hold x_1 constant
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- f_{22} : 2nd derivative w.r.t. x_2 , hold x_1 constant

Partial derivative recap: notation

- f_1 : derivative w.r.t. x_1 , hold x_2 constant
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- f_{11} : 2nd derivative w.r.t. x_1 , hold x_2 constant
- f_{22} : 2nd derivative w.r.t. x_2 , hold x_1 constant
- f_{12} : derivative w.r.t. x_1 , hold x_2 constant. Then derivative w.r.t. x_2 , hold x_1 constant
 - $f_{12} = f_{21}$, order doesn't matter

Partial derivative recap: example

$$f(x_1, x_2) = -3(x_1 - 2)^2 - (x_2 - 3)^2$$

Partial derivative recap: example

$$f(x_1, x_2) = -3(x_1 - 2)^2 - (x_2 - 3)^2$$

$$f_1 = (2)(-3)(x_1 - 2) - 0 = -6x_1 + 12$$

$$f_2 = 0 - (2)(x_2 - 3) = -2x_2 + 6$$

Partial derivative recap: example

$$f(x_1, x_2) = -3(x_1 - 2)^2 - (x_2 - 3)^2$$

$$f_1 = (2)(-3)(x_1 - 2) - 0 = -6x_1 + 12$$

$$f_2 = 0 - (2)(x_2 - 3) = -2x_2 + 6$$

$$f_{11} = -6$$

$$f_{22} = -2$$

$$f_{12} = f_{21} = 0$$

Lagrange multiplier method

- In econ we often optimize functions subject to (s.t.) a **constraint**

$$\max_{x_1, x_2} f(x_1, x_2) \quad \text{s.t.} \quad g(x_1, x_2) = 0$$

- Maximize the function over x_1 and x_2 given some condition for the values of x_1 and x_2
- No need to worry about SOC here, only FOC

Lagrange multiplier method

- Re-write the problem as a “Lagrangian”

$$\mathcal{L} = f(x_1, x_2) + \lambda g(x_1, x_2)$$

- The new term λ is called a Lagrange multiplier
 - Just think of λ as a third variable for now
 - $\mathcal{L}(x_1, x_2, \lambda) = f(x_1, x_2) + \lambda g(x_1, x_2)$
- Take 3 partial derivatives and set them equal to 0, then just algebra

Lagrange example: set-up

$$\begin{aligned}\max_{x_1, x_2} f(x_1, x_2) &= \frac{1}{4} \ln(x_1) + \frac{3}{4} \ln(x_2) \\ \text{s.t. } p_1 x_1 + p_2 x_2 &= \mathcal{I}\end{aligned}$$

$$\mathcal{L}(x_1, x_2, \lambda) = \frac{1}{4} \ln(x_1) + \frac{3}{4} \ln(x_2) + \lambda(\mathcal{I} - p_1 x_1 - p_2 x_2)$$

Lagrange example: set-up

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$$\mathcal{L}(x_1, x_2, \lambda) = \frac{1}{4} \ln(x_1) + \frac{3}{4} \ln(x_2) + \lambda(\mathcal{I} - p_1 x_1 - p_2 x_2)$$

FOC:

$$\frac{\delta \mathcal{L}}{\delta x_1} = 0$$

$$\frac{\delta \mathcal{L}}{\delta x_2} = 0$$

$$\frac{\delta \mathcal{L}}{\delta \lambda} = 0$$

Lagrange example: solution

FOC:

$$\frac{\delta \mathcal{L}}{\delta x_1} = \frac{1}{4} \frac{1}{x_1} - \lambda p_1 = 0 \implies \frac{1}{4x_1} = \lambda p_1$$

$$\frac{\delta \mathcal{L}}{\delta x_2} = \frac{3}{4} \frac{1}{x_2} - \lambda p_2 = 0 \implies \frac{3}{4x_2} = \lambda p_2$$

$$\frac{\delta \mathcal{L}}{\delta \lambda} = \mathcal{I} - p_1 x_1 - p_2 x_2 = 0$$

We can get rid of λ by dividing the 1st equation by the 2nd equation:

$$\frac{\frac{1}{4x_1}}{\frac{3}{4x_2}} = \frac{\lambda p_1}{\lambda p_2}$$

$$\frac{x_2}{3x_1} = \frac{p_1}{p_2}$$

$$x_2 = 3x_1 \frac{p_1}{p_2}$$

Lagrange example: solution

Finally, plug the expression for x_2 into the 3rd equation to solve for x_1^* :

$$x_2 = 3x_1 \frac{p_1}{p_2}$$

$$\mathcal{I} = p_1 x_1 + p_2 \left(3x_1 \frac{p_1}{p_2} \right)$$

$$\mathcal{I} = p_1 x_1 + (3x_1 p_1)$$

$$\mathcal{I} = p_1 (x_1 + 3x_1)$$

$$\frac{\mathcal{I}}{p_1} = 4x_1$$

$$\frac{\mathcal{I}}{4p_1} = x_1^*$$

Lagrange example: solution

We already found that $x_2 = 3x_1 \frac{p_1}{p_2}$, so just plug in x_1^*

$$x_2 = 3 \left(\frac{\mathcal{I}}{4p_1} \right) \frac{p_1}{p_2}$$

$$x_2^* = \frac{3\mathcal{I}}{4p_2}$$

$$x_1^* = \frac{\mathcal{I}}{4p_1}$$